

Jason Wakim Richards

+44 77 08 348 077 | jason2richards@gmail.com

github.com/richarJ2002 | linkedin.com/in/jason-richards-jwr2002/ | richarj2002.github.io/personal_website/

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SpaceX

Starfall Guidance, Navigation, and Control Team
Hawthorne, California, USA

Application for GNC Engineer (Starfall)

Dear Hiring Committee,

I am applying for the GNC Engineer (Starfall) position. My background combines a First-Class MEng in Aerospace Engineering, current doctoral research in autonomous UAV systems, and hands-on development across spacecraft ADCS, planetary-rover GNC, and real vehicle autonomy. Starfall's expectation that GNC engineers own algorithms, software, simulation, analysis, and vehicle integration matches how I work: I develop and debug the code, integrate it with sensors and hardware, analyze performance, document the design, and carry the system through testing. The opportunity to apply that approach to an entry capsule supporting logistics and in-space manufacturing is highly motivating.

At Airbus Defence & Space, I developed C/C++ software on Linux for an offline GNC simulation, coupling a natural-scene generator to produce synthetic imagery and LiDAR scans for algorithm development and evaluation. I also led LiDAR test activities during Integrated BreadBoarding 3 and 4: I designed a networked ROS sensor test bench, integrated it with rover breadboards, executed test procedures, and collected and analyzed campaign data. The campaigns supported an end-to-end Sample Fetch Rover concept-of-operations demonstration at TRL 6. This gave me practical experience connecting simulation models, GNC software, sensors, vehicle interfaces, performance data, and time-constrained operations in a multidisciplinary Agile team.

As ADCS Lead for the JamSail 3U CubeSat, I translated the mission concept of operations into requirements, architecture, interfaces, and a verification plan. I developed embedded C flight software and a C/C++/CMake simulation of rigid-body dynamics, attitude estimation, and control across mission modes. The determination chain combined TRIAD with an EKF fusing sun-sensor, magnetometer, and IMU measurements, supported by sensor filtering and a C implementation of the IGRF environmental model. This work developed the state estimation, attitude-control, spacecraft-systems, and simulation-verification foundation central to Starfall GNC.

I now develop and deploy C++ and Python autonomy software on physical UAVs using ROS 2, PX4, embedded Linux, camera/IMU sensing, and OpenVINS visual-inertial odometry. My work spans estimation, semantic mapping, planning, control, simulation, laboratory integration, flight testing, and performance analysis. I troubleshoot across software, sensing, estimation, and vehicle interfaces and build reproducible Docker, calibration, version-control, and technical documentation workflows. My research also includes shared semantic maps and mission-aware multi-vehicle coordination, directly complementing Starfall's interest in fleet-level logistics and recovery operations.

Although my vehicle experience has focused on spacecraft ADCS, planetary rovers, and UAVs rather than atmospheric-entry capsules, the underlying work—dynamics, multisensor estimation, attitude control, trajectory planning, GNC simulation, performance evaluation, and reliable C/C++ integration—is strongly transferable. I can contribute immediately in those areas while rapidly building deeper expertise in entry guidance, deorbit and trajectory optimization, and atmospheric, aerodynamic, and thermal modeling. I would welcome the opportunity to bring this combination of software ownership, systems judgment, and hands-on testing to the Starfall GNC team.

Yours sincerely,

Jason Wakim Richards